# ENGINEERING SCIENCE

Theory booklet

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# Programming

## Topic 1: Electronic control systems

## Topic 2: Microcontrollers

## Topic 3: Programming flowcharts

## Topic 4: Programming PBasic - missing

## Learning Intentions

- Gain the ability to design and evaluate solutions to engineering problems in a range of contexts.
- I will gain knowledge and understanding of key concepts related to electronic and microcontroller-based systems, and their application.
- To know what a microcontroller is and how it's used.
- I will know advantages and disadvantages of microcontroller based control systems compared to a hard-wired electronic equivalent.

## Success Criteria

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it and ask for help.
Amber: I need a bit of help to fully understand this topic.
Green: I am confident in my understanding of this topic.

- ☐ I can explain what a microcontroller is, how it works and can give examples of where they are used.
- ☐ I can state the advantages & disadvantages of using control systems in industry.
- ☐ I can explain the following terms; RAM, ROM, ALU, EEPROM, I/O port, bus and clock.
- ☐ I can identify and use various input and output transducer devices.
- ☐ I can use correct symbols (start, stop, input, output, branch, loop) to construct flowcharts showing solutions to simple control programs, involving time delays, continuous and fixed loops.
- ☐ I can use suitable commands, including high, low, for…next, if…then, pause, end (or their equivalents) to construct programs to solve simple control problems, involving time delays, continuous and fixed loops.
- ☐ I can use the computer and stamp controller to test my programs.

## Electronic Control Systems

Many electronic devices have been developed to make life easier, safer, to help with work and for entertainment purposes. Some devices are purely electronic devices (for example a digital watch) however many of these devices also control mechanisms (for example the eject mechanism in a video recorder) and so can be described as mechatronic devices.

Both electronic and mechatronic devices all have one thing in common they are all electronic control systems, which you learned about in unit 2.

## System diagram

Below is a universal system diagram. It identifies only 3 things which makes a system operate.




A SUB-system diagram is more detailed as it unpacks what is actual happening inside the 'process' box. It also shows the real world input signal (for example light and heat) and the real world output signals (for examples movement or sound).

The Universal System diagram for a warning device for a freezer in a restaurant would be drawn as shown below:




Or to put it another way:




**Input transducers** are electronic devices that detect changes in the 'real world' and send signals into the process block of the electronic system.

**Output transducers** are electronic devices that can be switched on and off by the process block of the electronic system.

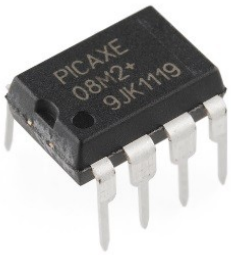
## Programmable control

Programmable control concerns an electronics engineer. It is about programming microcontrollers, which are chips that do a range of jobs needed in our modern world. Depending on its programming, a microcontroller can control simple electronic toys, complex manufacturing processes in industry, and anything in-between.

Microcontrollers are used very commonly and their use has drastically increased recently, as they are used in almost every electronic device. They are included in household and everyday items, including washing machines, microwave ovens and smartphones. In industry they are used in machine automation, process control production and conveyor lines. This has resulted in lower costs for companies and increased production, and has improved the consistency of quality. This has caused modern consumers to expect high

quality when they are purchasing and using manufacture goods. To engineers, these are the challenges that make electronic engineering interesting.

## PICAXE




PICAXE is a type of microcontroller that can be programmed using PBasic. They work in a very similar way to Ardunio or BBC microbits, however they use a different programming language. This course can be taught with either microcontroller, but we favour PBasic as we can use Yenka to program the PICAXE chips.

Once you have produced the flowchart for any given problem, you will then work out the coding, and this must be done in the correct language for the microcontroller you are using.

# The Microcontroller

A micro controller is often described as a computer on a chip, or a very small CPU. Microcontrollers have a controller and memory built into a single chip. They are used for sensing inputs from the real world and controlling outputs devices based on these inputs. As they are small and inexpensive they can easily be built into other devices to make these products more intelligent and easier to use.

Microcontrollers are usually programmed for a specific electronic product. For instance, a microwave oven may use a single microcontroller to process information for all its electronic devices (clock, buttons, dials, turntable, microwave emitter). By altering the microcontroller program, the same 'brand' of chip can do many different tasks.

## Advantages of using microcontrollers over hard wired circuits are:

- Increased reliability, as there are fewer components to break
- Reduced quantity of stock, as one microcontroller replaces several parts
- Reduces the cost of devices as there are fewer components
- Reduces the size of devices due to fewer components
- Simplified product assembly, so it is quicker to assemble devices and smaller end products
- Greater product flexibility and adaptability since features are programmed into the microcontroller and not built into the electronic hardware. This means a user can adjust or personalise the program.
- Rapid product changes or development by changing the program and not the electronic hardware
- Fewer natural resources used and less fumes produced, due to lower component numbers needed
- Can be reprogrammed/reusable therefore less waste (to landfill)

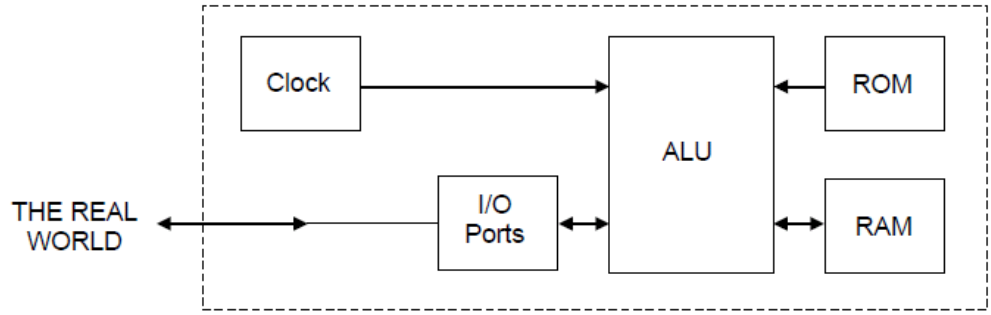
## Disadvantages of using microcontrollers

- The need software and access to a computer to be programmed
- Software may be expensive and need updated frequently
- They are often more expensive than other ICs

# Inside a Microcontroller

Although microcontrollers are relatively cheap, they are very complex devices containing thousands of transistors, resistors and other electronic components. All this is packaged inside the PICAXE chips you will use.

The main features of the micro controller are shown in the sub system diagram below.




*ROM*

The ROM (read only memory) contains the operating instructions for the micro controller. The ROM is 'programmed' before the micro controller is installed in the target system, and the memory retains the information even when the power is removed.

*RAM*

The RAM (Random-Access Memory) is 'temporary' memory used for storing information whilst the program is running. This is normally used to store mathematical answers that the micro controller comes out with as it is working. This memory is 'volatile', which means that as soon as the power is disconnected the contents of the memory are lost.

*ALU*

The processing unit (Arithmetic and Logic Unit) is the 'control centre' of the micro controller. It operates by reading instructions from the ROM and then carrying out the mathematical operations for each instruction.

*Clock*

The clock circuit controls the speed at which these operations occur. The clock circuit within the micro controller 'synchronises' all the internal blocks (ALU, ROM, RAM, etc.) so that the whole system works correctly.

*Buses*

Information is carried between the various blocks of the micro controller along 'groups' of wires called buses. The 'data bus' carries data between ALU and RAM and the 'program bus' carries the program instructions from the ROM to the ALU.

*EEPROM*

EEPROM (Electrically Erasable Programmable Read Only Memory) is a type of memory that can be reprogrammed when desired, but it also keeps the program when the power supply is removed. This means the microcontroller will start to run the program currently in the memory whenever the power supply is connected.

*I/O port*

(Input/Output port) is a socket on a computer that a cable is plugged into. The port connects the CPU to a peripheral device via a hardware interface or to the network via a network interface. Keyboard, mice and memory sticks use I/O ports.

## Microcontroller pin connections

To clearly communicate the input and output connections a microcontroller has, questions will often give you a table with pin connection details. You must be able to interpret these tables and understand how they link the microcontroller to other electronic components.

| Input Connection | Pin | Output Connection |
|---|---|---|
| | 7 | Display board |
| | 6 | LED |
| | 5 | Buzzer |
| Start switch | 1 | |
| Question switch | 0 | |

This example shows a microcontroller that has two input connections (the question switch to pin 0, and the start switch to pin 1) and three output connections (the buzzer to pin 5, the LED to pin 6 and the display board to pin 7).

When controlling an output, the pin must be mentioned if known. If a pin is on, it is known to be HIGH or 1. If a pin is off it is known to be LOW or 0.



Answer the following in your jotter.

a) List the advantages of using a microcontroller in product design. Try to do this without looking back at the list of advantages in this booklet to check your knowledge.

b) List the dis-advantages of using a microcontroller in product design. Try to do this without looking back at the list of advantages in this booklet to check your knowledge.

c) State three devices that may contain a microcontroller. Explain why it is an advantage for these products to use a microcontroller.

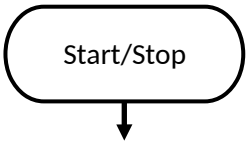
d) Within a microcontroller is six components (main features). For each feature, explain its purpose.

# Programming Flowcharts

Flowcharts are commonly used to explain how a program works. As flowcharts are drawn graphically they often make programs easier to understand. A flowchart should be drawn for each program you develop. They use different symbols to represent the different pasts of a program.

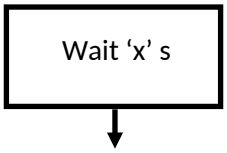
## *Start/stop symbol (terminator symbol)*

The start/stop symbol shape is a rectangle with rounded ends. Each flow chart must contain **one** start and usually **one** stop symbol unless it is a continuous loop.




## *Wait symbol (process symbol)*

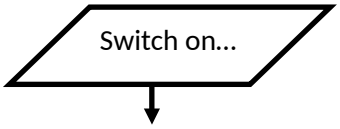
The 'wait' symbol is a rectangle. The text inside the symbol explains how long the time (x) delay is its numerical value.




Time in seconds must be recorded with the abbreviation 's' and **not** 'secs'

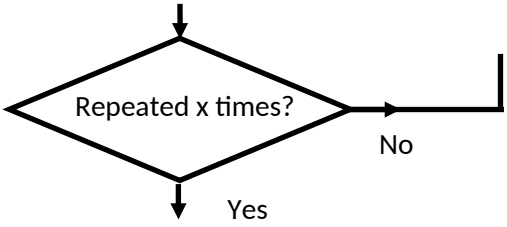
## *Input/Outputs symbol*

The 'outputs' symbol is a parallelogram. The text inside the symbol explains which output pins are switched on or off at any time.

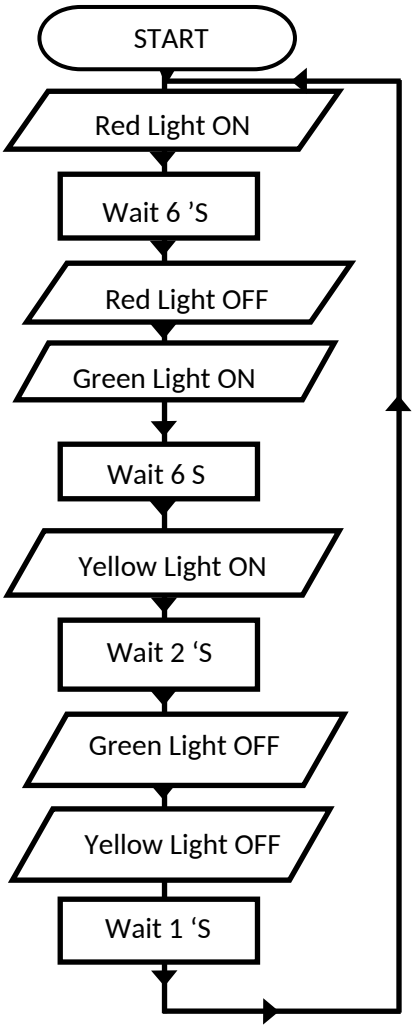



## *Decision Box*

The Decision box is a diamond. The program uses this to check if something has been completed.



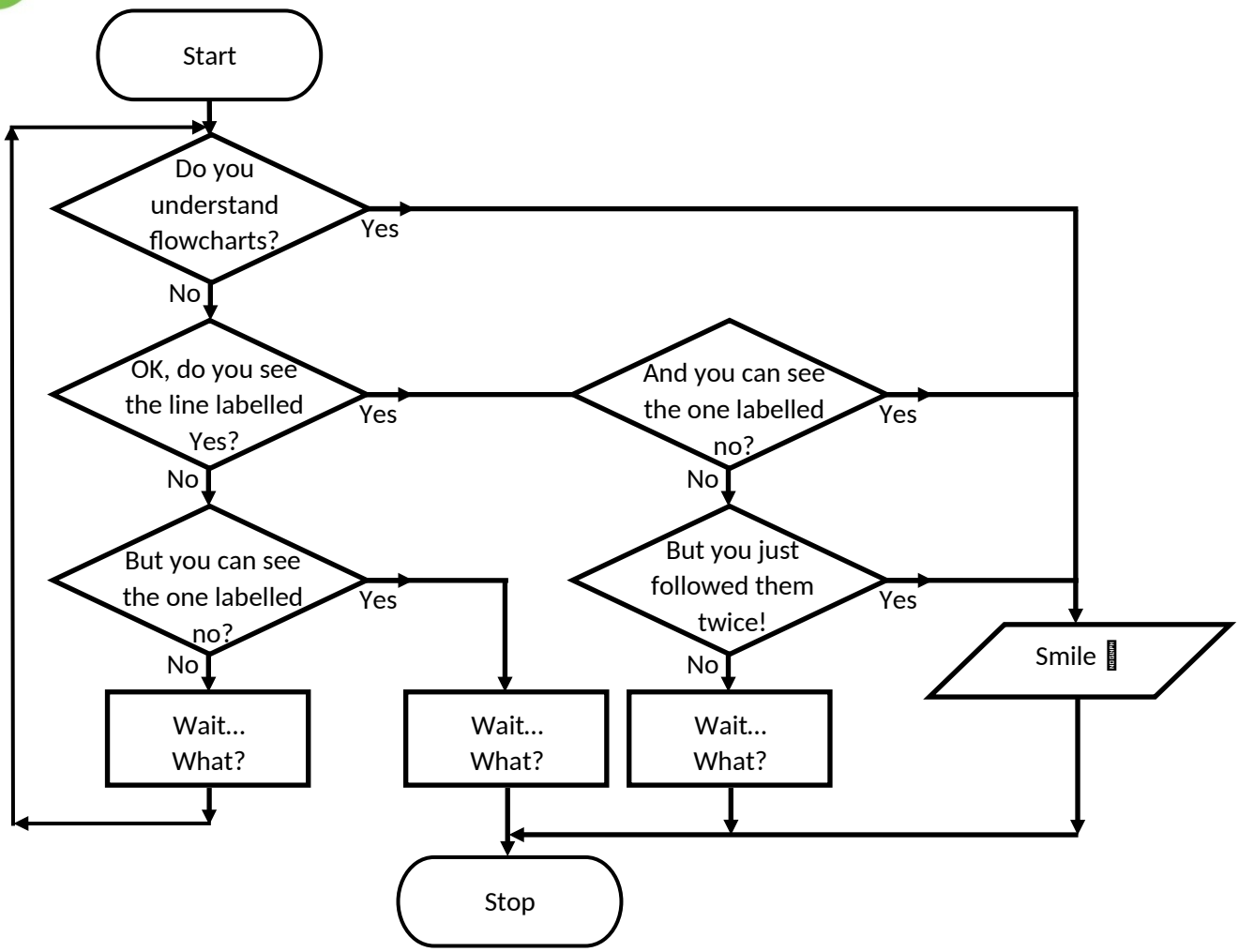

## *Example of a continuous loop*




## *Continuous loop*

Sometimes it is necessary to create programs that loop' forever', as is the case with traffic lights. There is no 'stop' symbol because the program never ends! An example is down the right of this page. You can also have a decision loop within a flowchart diagram. On the path line, there must be an arrow to identify the direction of decision which to follow.

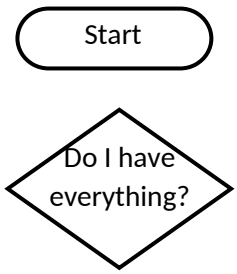
2. Take your time to make sure you understand flowcharts. You must know how to read and write them. Test your understanding with the flowchart below.




This flowchart shows that we can model many activities accurately with flowcharts. They are fundamental when it comes to programming and all electronics engineers plan the programs they write using flowcharts before transforming them into their programming language of choice.

It is possible for us to build and simulate flowcharts in Yenka, however we **must always plan them first**.

3. Write out the flowchart for making a cup of tea to your own taste. Another decision command should be used to identify if the water has boiled. Here is an example of how to start:




4. Write a program for a hot drinks machine. It has an option of making 4 hot beverages. Tea, coffee, hot water and a hot chocolate. However, you must also allow changes to the

drinks. Does it have sugar? Does it have milk? Remember you have to tell the system to stop these milk and sugar commands as well as start them (how long is the milk being added?).

5. A set of temporary traffic lights is required for a system of road works. Draw a flow chart for the lights sequence shown above by one set of traffic lights. Use the times given in the table.

| | |
|---|---|
| RED | 10 's |
| RED and AMBER | 2 's |
| GREEN | 10 's |
| AMBER | 2 's |

## Fixed loops in flowcharts

Sometimes we need a task to be completed a given number of times before the program ends. This is known as a fixed loop and an example is shown. Sometimes fixed loops are called finite loops, and in programming terms they are know as 'for...next' loops, which will be covered later in programming PBasic.

### *Worked example*

A toy robot is controlled by a microcontroller and follows the sequence shown.

1. A start switch must be activated for it to switch on
2. The legs start moving
3. After 2 seconds, the LED's on the gun light up
4. After 2 seconds, the arms move
5. After 5 seconds the arms stops and the LED's go off
6. The legs then stop moving
7. Septs 2-6 repeat three times
8. The sequence then repeats

| Input connection | Pin | Output connection |
|---|---|---|
| | 3 | Legs |
| | 2 | Arms |
| | 1 | LEDs |
| Start switch | 0 | |

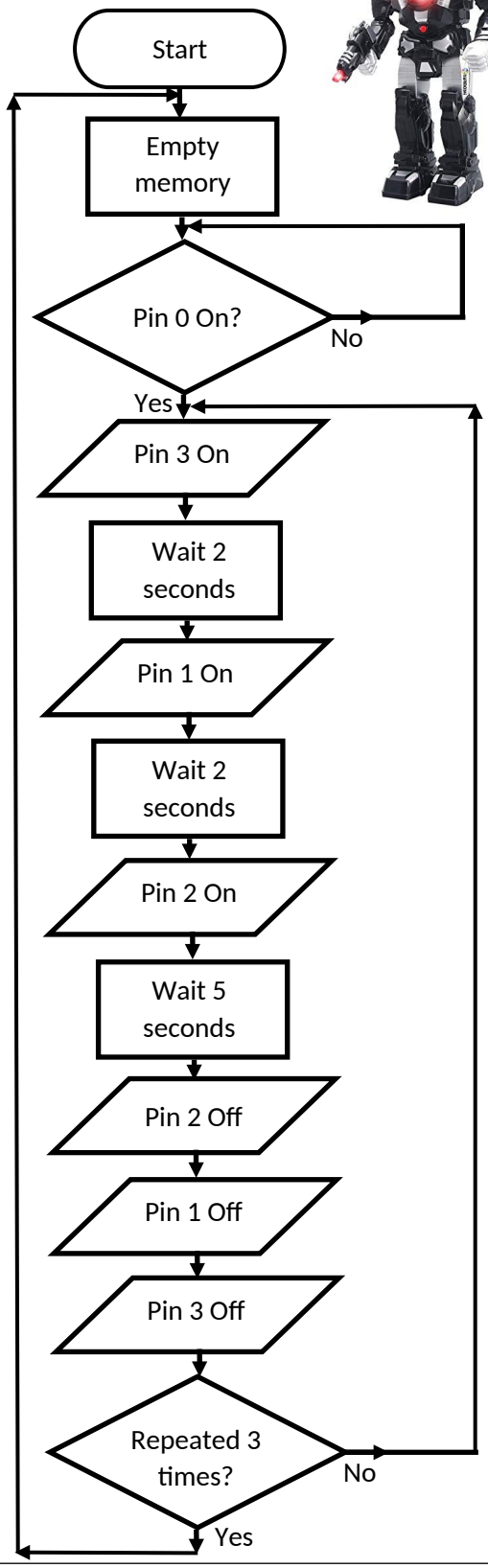
Note:

'HIGH' command also means 'on' (1)

'LOW' command also means 'off' (0)

Please use ON and OFF for any system control.





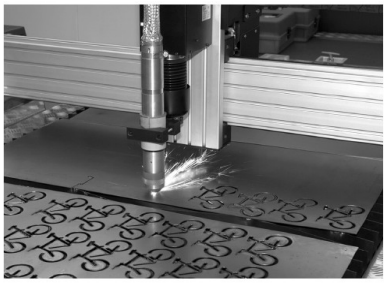
6. A microwave operates with the following sequence. This time, replace items like the turntable on with 'Pin 2 on'. You may want to start using the pin diagram below to identify within the microcontroller. Draw a flowchart for this sequence

| | |
|---|---|
| 1. | Light on |
| 2. | Turntable on |
| 3. | Magnetron on |
| 4. | Wait 30's |
| 5. | Magnetron off |
| 6. | Wait 10's |
| 7. | Turntable off |
| 8. | Buzzer on |
| 9. | Wait 0.5's |
| 10. | Buzzer off |
| 11. | Light off |

| Input connection | Pin | Outcome connection. |
|---|---|---|
| | 0 | |
| | 1 | Light |
| | 2 | Turntable motor |
| | 3 | Buzzer (sound) |
| | 4 | Magnetron (microwaves) |

Your next task is to build flowcharts in Yenka. Your teacher will show you how to model PICAXE connections and how to create linked flowcharts that simulate a running program. (In school only)

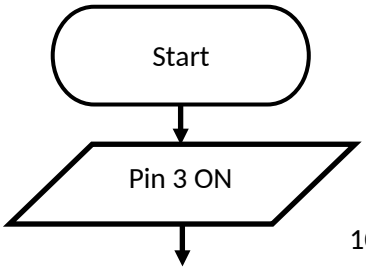
7. A laser cutting machine is used to cut sheet steel, as shown in the photograph. The laser is positioned by motors A and B which are operated by a microcontroller. Input and output connections to the microcontroller are shown in the table below.



| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Motor A forward (move right) |
| | 6 | Motor A reverse (move left) |
| | 5 | Motor B forward (move forward) |
| | 4 | Motor B reverse (move back) |
| | 3 | Laser on |
| | 2 | |
| | 1 | |
| | 0 | |

The cutter is required to perform the following sequence of operations:

1. Switch on the laser
2. Move right for 0·5 seconds
3. Move forward for 0·5 seconds
4. Repeat steps 2 and 3 four times
5. Switch off laser and motors



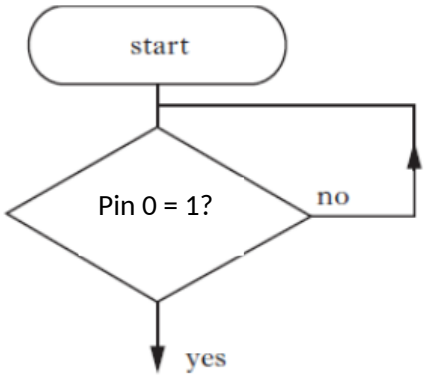
Complete the system flowchart to produce the required sequence of operations. The flowchart must include appropriate pin numbers. You may use information from the National 4/5 Data Booklet provided.

8. A car- park barrier uses a microcontroller to operate. Complete the flowchart for the barrier operation. The operation of the system is shown below.

- When the button is pressed, the machine will eject the parking ticket.
- Once the ticket is taken, it will raise the barrier for 5 seconds.
- When it has sensed the car has moved away it will lover the barrier for 5 seconds.
- The sequence will repeat.

9. A railway uses a microcontroller to operate a barrier. The operation of the system is shown below. Complete the flowchart for the barrier operation which has been started for you.

- When a train passes over a magnetic switch on the track, the barrier lowers and two red warning lights flash.
- The lights continually flash for 2 seconds on then 1 second off. When one light is one, the other is off.
- A barrier limit switch is positioned in both directions. Once the limit switch has been pressed by the barrier being fully open or closed, the barrier must stop.
- Once the train has passed the crossing and a second magnetic switch has been activated, the barrier will rise and the lights turn off. Sequence will then repeat.




| Input connection | Pin | Output connection |
|---|---|---|
| First magnetic sensor | 0 | |
| Second magnetic sensor | 1 | |
| Open limit switch | 2 | |
| Closed limit switch | 3 | |
| | 4 | Barrier motor raise (anti clockwise motion) |
| | 5 | Barrier motor lower (clockwise motion) |
| | 6 | Red light 1 |
| | 7 | Red light 2 |

10. A water jet uses high-pressure water to cut metals, such as sheet steel. The position of the water jet is controlled by motors A and B, which are operated by a microcontroller. The input and output connections to the microcontroller are shown in the table.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Motor A forward (clockwise motion) x-axis |
| | 6 | Motor A reverse (anti clockwise motion) |
| | 5 | Motor B right (clockwise motion) y-axis |
| | 4 | Motor B left (anti clockwise direction) |
| | 3 | Water jet on |
| | 2 | |
| | 1 | |
| | 0 | |

The cutter is required to perform the following sequence of operations:

1. Switch on the water jet.
2. Move right for 1.5 seconds.
3. Move forward 3.5 seconds.
4. Repeat steps 2 and 3 five times.
5. Switch off the water jet and the motors

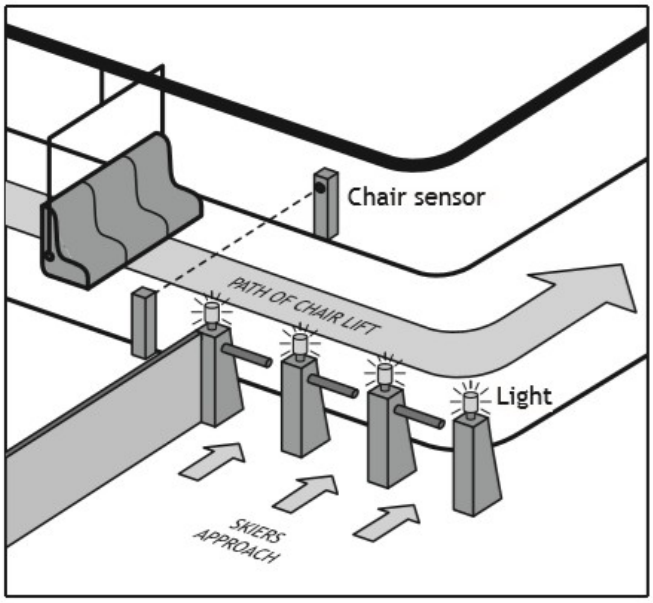
Draw the flowchart for this sequence. The flowchart must include appropriate pin numbers.

Extension: Model your flowcharts answers from Q9 and Q10 in Yenka. Use associated PICAXE chip 18M2.

Pass paper N5 2014 Q13

11.

A chair lift system at a ski resort is operated by a microcontroller.



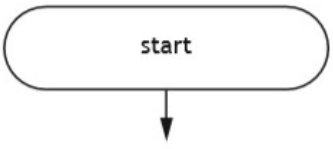

The system must perform the following sequence.

- When the chair sensor is triggered warning lights flash 4 times over a 2 second period.
- The barriers will then open.
- The system will pause for 3 seconds to allow the skiers through.
- The barriers will then close.
- The sequence will repeat.

Input and output connections to the microcontroller are shown in the table below.

| Input Connection | Pin | Output Connection |
|---|---|---|
| | 2 | Barriers (1 = open) |
| | 1 | Warning lights |
| Chair sensor (1 = chair sensed) | 0 | |

In your jotter, complete the flowchart for the control sequence with reference to the Data booklet and input/output connections. Include all pin numbers. The first flowchart command is shown below.



## Extension questions

1. Switch on a red LED for 1 second and switch red LED off for 1 second then stop the program.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | |
| | 6 | |
| | 5 | Red LED |
| | 4 | |
| | 3 | |
| | 2 | |
| | 1 | |
| | 0 | |

2. Switch on red and green LEDs for 2 seconds and switch Red and green LEDs off for 1 second then loop the program.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Green LED |
| | 6 | |
| | 5 | Red LED |
| | 4 | |
| | 3 | |
| | 2 | |
| | 1 | |
| | 0 | |

3. Repeat the following sequence 3 times and then stop the program: switch on a red LED for 2 seconds and switch the red LED off for 2 seconds. Use the pin and table from Q1.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Green LED |
| | 6 | |
| | 5 | |
| | 4 | Red LED |
| | 3 | |
| | 2 | |
| Switch | 1 | |
| | 0 | |

4. If a switch is pressed, complete the following sequence then loop the program: turn on a red & green LEDs for 2 seconds and switch red & green LEDs off for 2 seconds.

5. If switch 1 **and** switch 2 are pressed, turn green on for 1s and off for 1s and stop the program.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Green LED |
| | 6 | |
| | 5 | |
| | 4 | Red LED |
| | 3 | |
| Switch 2 | 2 | |
| Switch 1 | 1 | |
| | 0 | |

6. If switch 1 is pressed, green on for 2s and off for 2s or if Switch 2 is pressed turn purple on for 1s and off for 1s. Repeat program for ever.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Green LED |
| | 6 | |
| | 5 | |
| | 4 | Red LED |
| | 3 | |
| Switch 2 | 2 | |
| Switch 1 | 1 | |
| | 0 | |

## Exam style questions

**Q15 2015**

A sign used to display a car's speed is shown below.



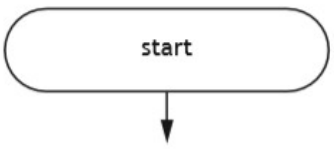

The sign is operated by a microcontroller. Input and output connections to the microcontroller are shown in the table below.

| Input connections | Pin | Output connections |
|---|---|---|
| | 7 | |
| | 6 | speed display |
| | 5 | "smiley" face display |
| | 4 | |
| | 3 | |
| speed sensor (1 = too fast) | 2 | |
| | 1 | |
| | 0 | |

The sign operates using the following sequence.

1 The speed sensor measures the speed of the car.
2 If the car is moving too fast, the speed display sign is switched on for 0·5 seconds.
3 If the car is not moving too fast, the "smiley" face display is switched on for 1 second.
4 The sequence then repeats.

In your jotter, complete the flowchart for the control sequence with reference to the Data booklet and input/output connections. Include all pin numbers. The first flowchart command is shown below. **6**




State the type of program loop used in the operation of the sign. **1**

**Q12 2016**

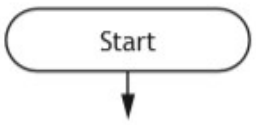
A microcontroller is used to operate the model in the sequence shown below.

1 When a start switch is activated the arms will move.
2 After 1·5 seconds the dinosaur will then roar for 1 second.
3 After another 0·5 seconds the arms should halt.
4 Steps 2 and 3 will repeat three times.
5 The sequence will then repeat.

Input/output connections to the microcontroller are shown in the table below.

| Input Connection | Pin | Output Connection |
|---|---|---|
| | 2 | arms |
| | 1 | roar |
| start switch | 0 | |

(c) Complete the flowchart for the control sequence with reference to the Data Booklet and input/output connections. Include **all** pin numbers. **8**



**Q15 2019**

A set of hair straighteners is shown.



A microcontroller is used to operate the hair straighteners using the following simplified sequence.

- When a push switch is operated the heating plates turn on.
- A buzzer will then sound for 0·8 seconds and then turn off.
- An LED will then repeatedly turn on for 0·3 seconds and turn off for 0·3 seconds until the correct temperature is reached.
- When the push switch is pressed again the heating plates turn off.
- The sequence will then repeat.

Input and output connections to the microcontroller are shown in the table below.

| Input connection | Pin | Output connection |
|---|---|---|
| | 7 | Heating plates |
| | 6 | Buzzer |
| | 5 | LED |
| Temperature sensor (1 = correct temperature) | 1 | |
| Push switch | 0 | |

In your jotter, complete the flowchart for this simplified sequence, with reference to the Data booklet and input/output connections. Include all pin numbers and delay unit in your flowchart. The first two flowchart commands are shown below. **9**

